

There are a total of four problems. You have to solve **all** of them.

Problem 1 (CO1): DFA and Regular Languages (10 points)

Let $\Sigma = \{0, 1\}$. Consider the following languages over Σ .

$$L_1 = \{w : \text{the length of } w \text{ is at most three}\}$$

$$L_2 = \{w : w \text{ starts and ends with different letters}\}$$

$$L_3 = \{w : \text{the length of } w \text{ is at least two}\}$$

Now solve the following problems.

- Give the state diagram for a DFA that recognizes L_1 . (2 points)
- Give the state diagram for a DFA that recognizes L_2 . (3 points)
- Give the state diagram for a DFA that recognizes L_3 . (2 points)
- Find a shortest string in $\overline{L_1} \cap L_3$. Here $\overline{L}$ denotes the complement of the language L i.e., $\overline{L} = \Sigma^* - L$. (1 point)
- If you were to use the "cross product" construction shown in class to obtain a DFA for the language $L_2 \cap L_3$, how many states would it have? (1 point)
- How many states does the smallest DFA for $L_2 \cap L_3$ have? (1 point)

Problem 2 (CO1): Regular Expressions (10 points)

Let $\Sigma = \{0, 1\}$. Consider the following pair of languages over Σ .

$$L_1 = \{w : \text{the length of } w \text{ is divisible by three}\}$$

$$L_2 = \{w : \text{every second letter in } w \text{ is a } 0\}$$

Now solve the following problems.

- Write down a regular expression for the language L_1 . (2 points)
- Write down a regular expression for the language L_2 . (2 points)
- Your friend wants a regular expression for the language $\overline{L_1} \cap \overline{L_2}$ where $\overline{L}$ denotes the complement of the language L i.e., $\overline{L} = \Sigma^* - L$. She wants your help. You tell her to make use of the fact $\overline{L_1 \cap L_2} = \overline{L_1} \cup \overline{L_2}$.
 - Write down a regular expression for the language $\overline{L_1}$. (2 points)
 - Write down a regular expression for the language $\overline{L_2}$. (2 points)
 - Using the fact above, write down a regular expression for the language $\overline{L_1} \cap \overline{L_2}$. (2 points)

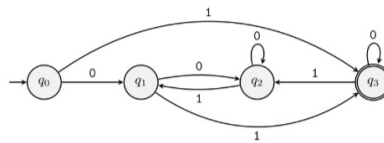
Problem 3 (CO1): Converting Regular Expressions to NFAs (10 points)

Convert the following regular expression over $\Sigma = \{a, b, c\}$ into an equivalent NFA. Note that $R_1 + R_2$ is the same as $R_1 \cup R_2$.

$$aa^* + (a + b^*(a + c)^*)^*$$

Problem 4 (CO1): Converting Finite Automata to Regular Expressions (10 points)

Convert the following DFA into an equivalent regular expression using the state elimination method. First eliminate q_2 , then q_3 and finally q_1 . You must show work.



After you are done with the test, please indicate where you stand on the smiley face spectrum.

